CHANDU PRASAD BHAIRAPU



AI Engineer

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github

ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in AI-driven products and full-stack web applications, from LLM pipelines and RAG to React frontends, Python backends and cloud deployment on AWS and Azure. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by close work with its domain experts.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the knowledge-extraction pipeline end-to-end, from data ingestion, chunking and prompt engineering to indexing, and making the architecture decisions between agentic, deterministic, and monolithic strategies; reporting directly to the CTO and working with SAP domain experts across the German energy sector.
- Developed Expert Interface, a graph curation web application (React frontend, Python microservices backend) deployed on Azure, giving domain experts a workflow to validate and approve knowledge-graph changes.
- Designed and optimized multi-agent extraction with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Built a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated into Claude Code; built with custom Claude Code skills and clean code principles.
- **Tech stack: Python, React, FastAPI, Azure, Docker, Kubernetes, Postgres, MongoDB, Neo4j, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, Vector Search, MCP**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Worked directly with business stakeholders to understand their decision-making workflows, advised where agentic AI could support them, and translated their requirements into a concrete solution approach.
- Built a full-stack web application with a React frontend and a Python FastAPI backend, using Postman for API testing, around a LangGraph agent that routes requests across vector database queries, general inference, and summarization.
- Containerized the stack with Docker and deployed it to AWS with CDK, following clean code practices, with version control and CI/CD in GitLab.
- **Tech stack: Python, React, FastAPI, Postman, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, LangGraph, LangChain**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated evaluation of answer quality and retrieval for the PrismaChat RAG system with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for the evaluation using CDK, and benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, AWS (Lambda, SQS, DynamoDB, CDK), Azure AI Search**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Deployed an interactive Streamlit dashboard exposing model predictions to chemists in real time, letting them simulate parameter changes and see predicted outcomes before running physical batches.
- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data, improving early identification of batches likely to deviate from target specification.
- **Tech stack: Python, Streamlit, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy and strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

PROJECTS

Project: Clinical Info Assistant | React | TypeScript | Python | FastAPI | LLMs

- Web application that answers questions about medicines from drug label texts, with separate user journeys for healthcare professionals and patients, cited sources on every answer, and a handoff when a patient question needs a doctor.
- React and TypeScript frontend on a Python FastAPI backend with retrieval and a grounding check; usage events and a funnel view to study user behavior and find friction in the journeys; clean, tested, maintainable code with pytest, UI tests and a labelled evaluation set run in CI.

TECHNICAL SKILLSET

- **Web Development:** Python, React, FastAPI, REST APIs, Python microservices, Streamlit, Postman (API testing)
- **AI/ML-Driven Products:** large language models, LLM APIs (LiteLLM), RAG, GraphRAG, agentic workflows (LangGraph), LangChain, MCP, prompt engineering, evaluation (RAGAS, PytrecEval), Hugging Face Transformers
- **Cloud & Delivery:** AWS (CDK, Lambda, EC2, SQS, S3, DynamoDB), Azure, Docker, Kubernetes, GitLab CI/CD, Git
- **Databases:** SQL, Postgres, MongoDB, Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Machine Learning:** Scikit-Learn, XGBoost, PyTorch, TensorFlow, time-series analysis
- **Ways of Working:** Agile, clean code, Claude Code (custom skills, agentic coding workflows)

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)